Lecture 11 - Regularized Optimization

ECE 447: Data Analysis and Machine Learning for Engineers

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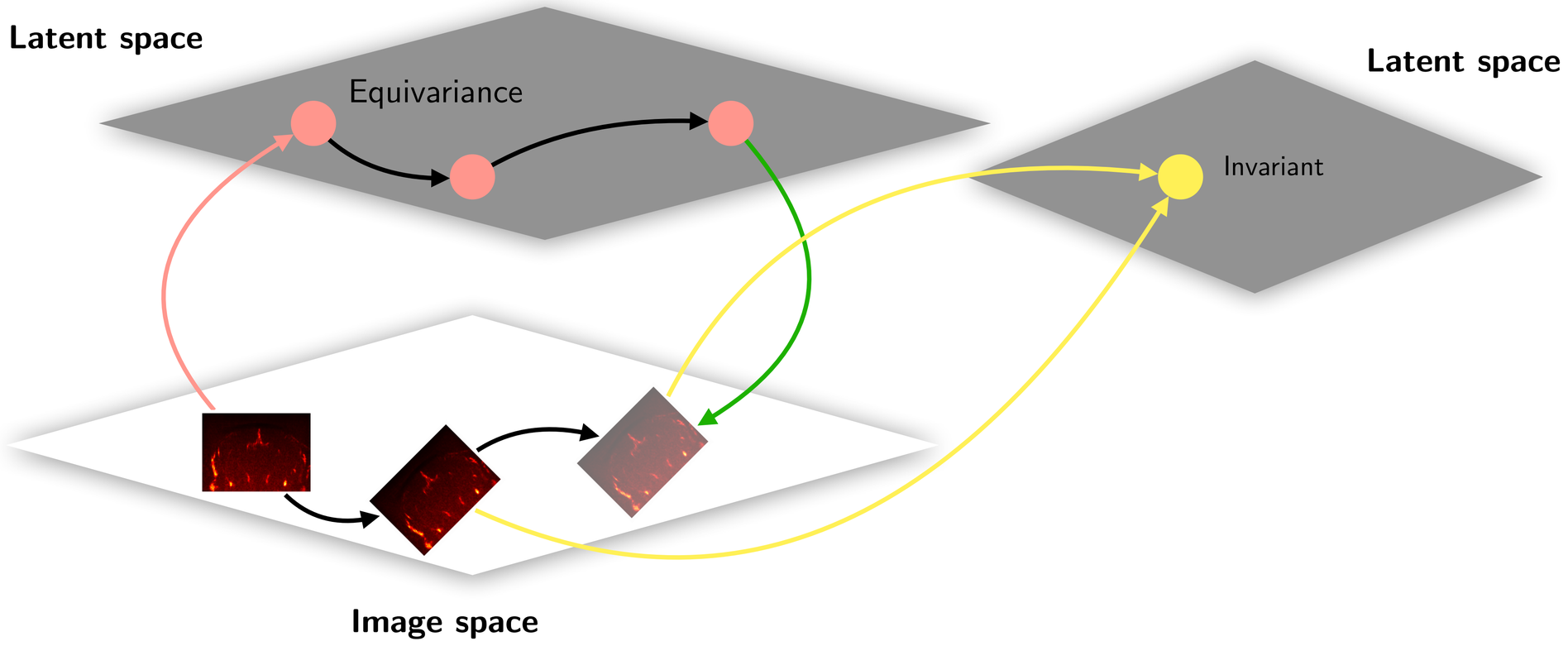
Notions of Invariance and Equivariance

This slide is informational, excluded from assessment.



This Lecture Covers ...

- Regularization
- Ridge Regression
- Lasso

Regularization in Optimization

Consider the optimization problem for linear regression

$$\min_{\boldsymbol{\theta}} \quad \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2$$

One can *regularize* the solution by adding an additional term

$$\min_{\boldsymbol{\theta}} \quad \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2 \; + \; \mathscr{R}(\boldsymbol{\theta})$$

Restrict the space of solutions.

High value of $\mathscr{R}(\boldsymbol{\theta})$ for undesired solution

Low value of $\mathscr{R}(\boldsymbol{\theta})$ for desired solution.

Why to Regularize?

- Promote a particular prior (structure) on the solution (e.g., smooth, low-rank, ...).

- Prevent overfitting.

- Improve generalization.

Ridge Regression — ℓ_2-norm Regularized Linear Regression

Given a set of data points $\{\boldsymbol{x}_i \in \mathbb{R}^p, y_i\}_{i=1}^n$, find the parameters $\boldsymbol{\theta}$ that minimizes:

$$\boldsymbol{\theta}_{\text{Ridge}} = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^p} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2 + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2 = (\boldsymbol{X}^\top\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$$

where $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad \boldsymbol{X} = \begin{bmatrix} -\boldsymbol{x}_1^\top- \\ -\boldsymbol{x}_2^\top- \\ \vdots \\ -\boldsymbol{x}_n^\top- \end{bmatrix} \in \mathbb{R}^{n\times p}.$

Useful regularization when the LS solution is not unique (e.g., $n \ll p$).

Ridge Regression — Gaussian Prior on Parameter

Consider the linear model

$$y_i = \boldsymbol{\theta}^\top \boldsymbol{x}_i + \epsilon_i \qquad , \epsilon_i \overset{iid}{\sim} \mathcal{N}(0,\sigma^2)$$

$$p(\boldsymbol{\theta}) \sim \mathcal{N}(0,\sigma_\theta^2 \boldsymbol{I})$$ *Prior on θ*

Goal: find the parameter that maximizes the posterior $p(\boldsymbol{\theta}|\boldsymbol{y})$.

likelihood ***prior***

posterior

$$p(\boldsymbol{\theta}|\boldsymbol{y}) = \frac{p(\boldsymbol{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{y})}$$

We are interested in maximum a posteriori (MAP) estimation to combine both likelihood and prior.

Ridge Regression — MAP Estimation

$$p(\boldsymbol{\theta}\,|\,\boldsymbol{y}) = \frac{p(\boldsymbol{y}\,|\,\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{y})}$$

$$\begin{aligned}
\boldsymbol{\theta}_{\text{MAP}} &= \arg\max_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad p(\boldsymbol{\theta}\,|\,\boldsymbol{y}) \\
&\quad \arg\max_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad \log p(\boldsymbol{\theta}\,|\,\boldsymbol{y}) \\
&\quad \arg\max_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad \log p(\boldsymbol{y}\,|\,\boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) \\
&\quad \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad -\log p(\boldsymbol{y}\,|\,\boldsymbol{\theta}) - \log p(\boldsymbol{\theta}) \\
&\quad \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2 + \frac{1}{2\sigma_\theta^2}\|\boldsymbol{\theta}\|_2^2
\end{aligned}$$

When you regularize your parameter with ℓ_2-norm, you have indeed assumed a Gaussian prior on the parameter.

Multiple by σ^2 and denote $\lambda = \frac{\sigma^2}{\sigma_\theta^2}$, we get the same objective as regularized LS.

Lasso - Least Absolute Shrinkage and Selection Operation

Lasso solves the following optimization problem for linear regression

$$\min_{\boldsymbol{\theta}} \quad \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2 \; + \; \lambda\|\boldsymbol{\theta}\|_1$$

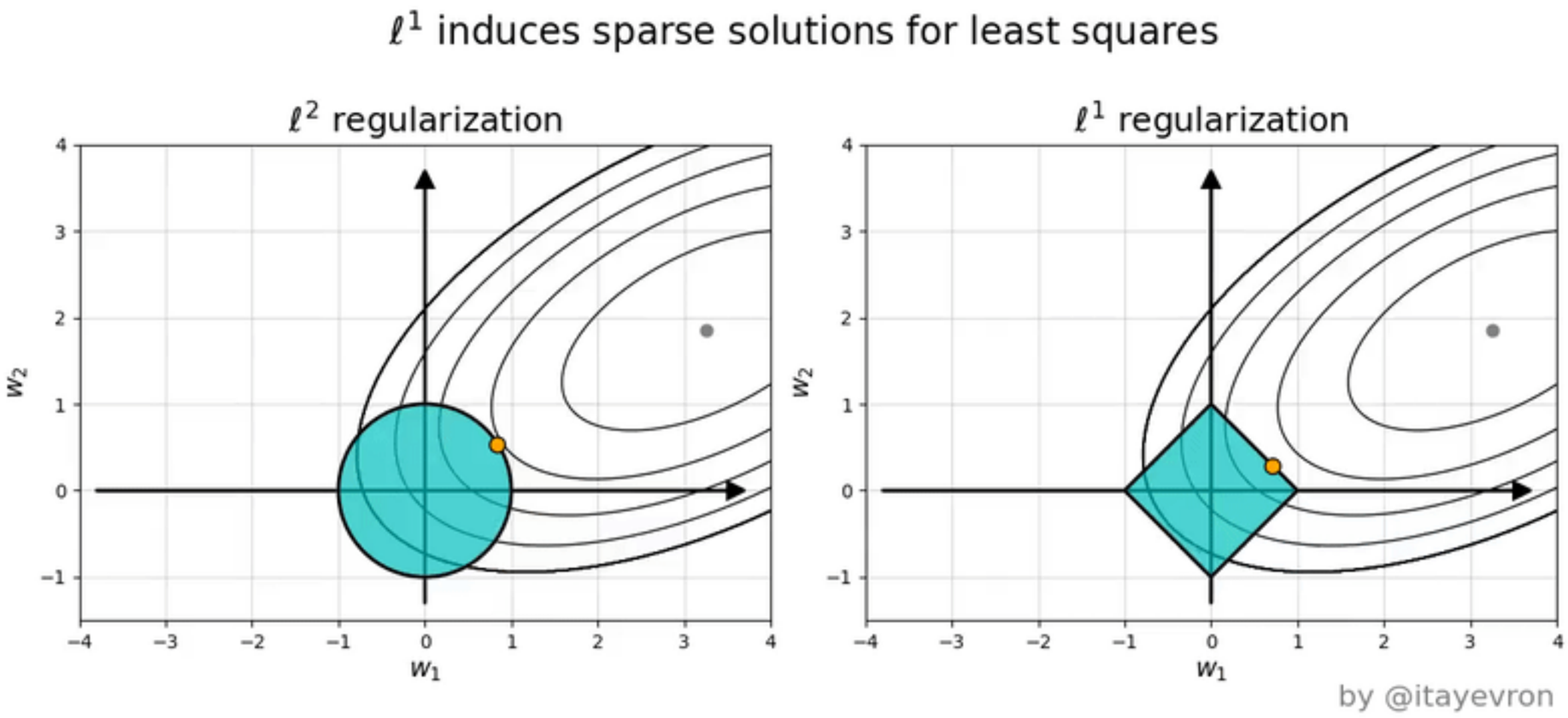
which regularize the ℓ_1-norm of the parameter.

In Bayesian, MAP perspective, what is the underlying prior on $\boldsymbol{\theta}$?

Laplace

$$p(\boldsymbol{\theta}) = \frac{\lambda}{2}\, e^{-\lambda\|\boldsymbol{\theta}\|_1}$$

What is the Effect of ℓ_2-norm and ℓ_1-norm on solution?



ℓ_2-norm: encourages parameter values to be small.

ℓ_1-norm: encourages parameter sparsity, i.e., small number of entries to be non-zero (active).

Takeaways

- *Regularizers shape the space of desirable solutions.*

- *Regularizers can be understood as* ***priors****.*